

Derivatives of Exponential and Logarithmic Functions

Section

3.3

A. Derivatives

$$1. \frac{d}{dx}(\log_a x) = \frac{1}{x \cdot \ln a}$$

$$2. \frac{d}{dx}(\ln x) = \frac{1}{x}$$

$$3. \frac{d}{dx}(a^x) = a^x \cdot \ln a$$

$$4. \frac{d}{dx}(e^x) = e^x$$

Examples: Find the derivative of each

1. (video) $f(x) = 5 \log_7 x$

2. (video) $f(x) = 17e^x + 5x - \sin x$

3. (video) $y = e^{-5 \tan x}$

4. (video) $y = e^{-5x} \cdot \cos(3x)$

5. (video) $f(x) = \ln x$

6. (video) $f(x) = 3x^6 \cdot \ln(3x^2 + 4)$

7. $f(x) = \ln(x^3 + 10)$

8. $f(x) = \ln\left(\sqrt{\frac{4x-6}{x+1}}\right)$

9. $f(x) = 5\sqrt{x} \cdot e^{2x}$

10. $f(x) = \log_3(5x^2 + 4)$

11. $f(x) = 7^{5x}$

B. Logarithmic Differentiation

Examples: Find the derivative of each

12. $y = x^{\sqrt{x}}$

13. $y = \frac{(9x+4)^7(x^3-5)^3}{\sqrt{3x-1}}$

